

CEGEP VANIER COLLEGE
Department of Computer Science
420-231-VA Database

Team Project- Deliverable #2: Database Schema

Deliverable 2 Due: April 10, 2026

Teacher: Nagat Drawel

Your deliverable must contain the following:

1. Descriptions of any changes made to **Deliverable 1** (in response to the feedback)
 - a. Please resubmit a modified version of the previous project deliverable showing the changes in your project's design, if any. If you do, please highlight the changed requirements. All deliverables must be kept consistent.
2. Normalize the relations to 3NF.
 - a. Explain your process and assumptions. In each step:
 - i. Show the complete schema at that level with the primary key(s) identified.
 - ii. Explain what functional dependencies are removed.
 - iii. List all existing functional dependencies.
3. Database Schema (SQL script file)
 - a. Create a database schema for your project using SQL DDL (Data Definition Language) statements. Choose appropriate data types for each attribute and include Primary Key and Foreign Key constraints, Check, Unique, and Not Null constraints. Please assume defaults for ON DELETE clauses for FOREIGN KEYS.
 - b. Explain why you chose your ON DELETE actions. As an example: *"if we delete a customer, then we delete their accounts"*).
 - c. Populate every relation with sufficient representative rows using DML (Data Manipulation Language) statements.
 - d. Create at least one SEQUENCE to generate values for one of the tables of your choice. Select the start value of your sequence and the increment by value. Do not cache any values, also do not cycle. You can add more sequences if you like. Also, at least one of your Primary keys should be defined as an **IDENTITY** property.

- e. Create two indexes to your database. Select the one that you think will be beneficial. It is not critical that these be the most important indexes, but the choices should make good sense in terms of the queries that you expect will be run on the database. Explain why you chose your indexes.
- f. Write at least two ALTER statements to either add/modify columns or add constraints to any of the tables.

What you should submit

1. You should submit SQL declarations of the relations and relations instances. Submit SQL scripts, containing all the DDL(s) and DML(s) needed for the database creation and the after mentioned requirements. This script file should be executed as a script using the selected DBMS. **In writing your DDL/DML statements, use a convention of CAPITALIZING all SQL reserved words and use indentation as necessary.**
2. You should submit a report (**PDF file**) that includes:
 - a. Changes from the Previous Deliverable
 - b. Normalization steps
 - c. Include a section named “Teamwork Summary”, in which the work done by each member must be clearly stated.
 - d. Besides the above, comment on the difficulties you and your group faced in doing this conceptual design task

Distribution of marks:

Topic	%
Changes in 1 st deliverable	20%
Normalization	20%
DDL Statements	20%
Create Indexes	10%
DML (Sample Test Data)	10%
Script File	5%
Individual Contribution	15%
Total	100%